14B 1 ACID–BASE TITRATIONS, pH CURVES AND INDICATORS

SPECIFICATION REFERENCE
14.15 14.16

LEARNING OBJECTIVES

- Be able to draw and interpret titration curves, using all combinations of strong and weak monoprotic and diprotic acids with bases, and apply these principles to diprotic acids and bases.
- Be able to select a suitable indicator for a titration, using a titration curve and appropriate data.

ACID–BASE TITRATIONS

END POINT AND EQUIVALENCE POINT

When you carry out a simple acid–base titration, you usually use an indicator to tell you when the acid and base are mixed in exactly the right proportions to react in equivalent amounts, as dictated by the stoichiometric equation. When the indicator changes colour, this is often described as the 'end point' of the titration.

The **equivalence point** is when the acid and base have reacted together in the exact proportions as dictated by the stoichiometric equation. When titrating an aqueous solution of a monobasic acid with an aqueous solution of a monoacidic base of the same concentration, 25 cm^3 of acid will react exactly with 25 cm^3 of base.

The pH at the equivalence point depends on the combination of acid and base used. For example, if you are titrating aqueous sodium hydroxide with dilute hydrochloric acid, then the pH at the equivalence point is 7.00 (at 298 K), as both the base and the acid are strong. The solution at the equivalence point will contain the salt sodium chloride.

If, however, ethanoic acid (a weak acid) is titrated against sodium hydroxide (a strong base), the solution at the equivalence point will contain the salt sodium ethanoate, and the pH will be greater than 7 (see **Section 14A.4**).

If hydrochloric acid (a strong acid) is titrated against aqueous ammonia (a weak base), the solution at the equivalence point will contain the salt ammonium chloride, and the pH will be less than 7.

LEARNING TIP

The term 'neutralisation point' should *not* be used to describe the point at which the acid and base have reacted in the exact proportions as dictated by the stoichiometric equation. As seen by the examples listed, the pH of the solution formed is not always 'neutral' (i.e. does not always have a pH of 7.00 at 298 K).

Remember that the term 'end point' refers to when the colour of the indicator just changes colour – this does not always occur at the equivalence point (see discussion of indicator choice later in this section).

TITRATION OF A STRONG ACID WITH A STRONG BASE

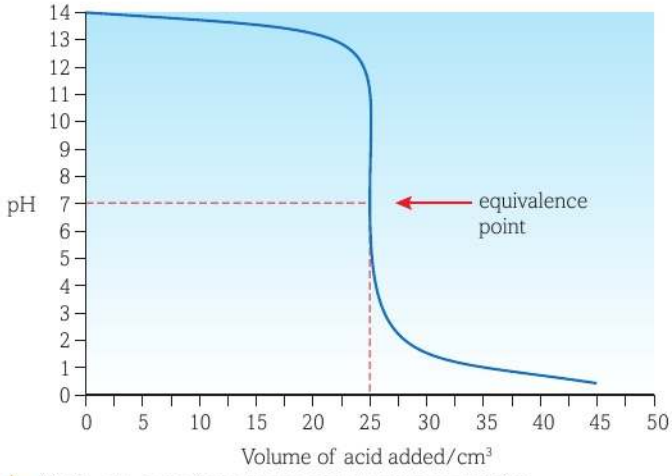
As you add an aqueous solution of an acid to an aqueous solution of a base, you might expect there to be a gradual change in the pH of the solution formed. This is not the case. When the pH of the solution is plotted against the volume of acid added, the shape of the curve depends on the nature of the acid and base used. The curves produced in this way are called 'pH titration curves'.



▲ **fig A** A pH meter.

We will start by looking at the pH titration curve when a strong acid is added to a strong base.

The following graph is produced when adding 1.00 mol dm^{-3} HCl(aq) to a 25 cm^3 sample of 1.00 mol dm^{-3} NaOH(aq). The pH is measured using a pH meter such as the one shown in **fig A**.



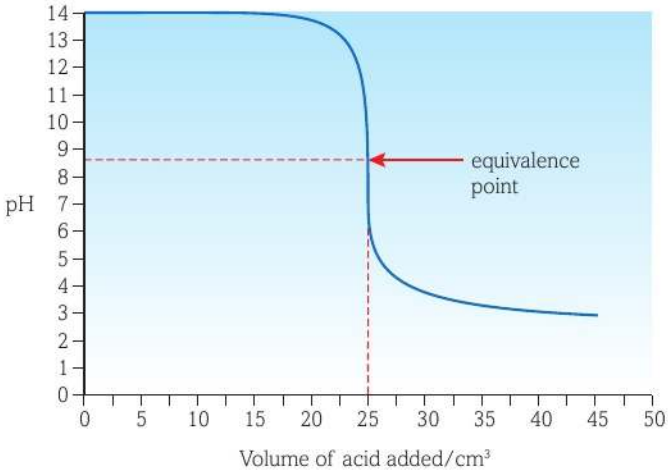
▲ **fig B** pH curve for a strong acid–strong base titration.

You can see in **fig B** that the pH falls only a very small amount until it is quite near the equivalence point. Then there is a really steep plunge. If you calculate the values, the pH falls all the way from 11.30 when you have added 24.90 cm^3 to 2.70 when you have added 25.10 cm^3.

There is a large 'steep section' to the curve. As we will see later in this section, this is an important point to consider when choosing an appropriate acid–base indicator to determine the end point of this titration.

TITRATION OF A WEAK ACID WITH A STRONG BASE

For this example we will add 1.00 mol dm^{-3} ethanoic acid to 25 cm^3 of 1.00 mol dm^{-3} sodium hydroxide.



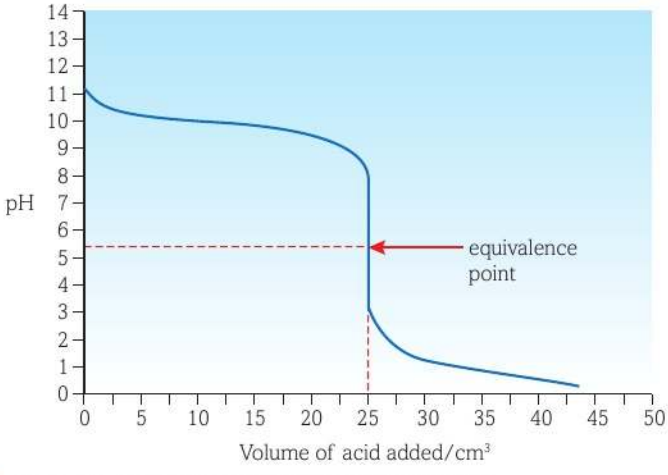
▲ **fig C** pH curve for a weak acid–strong base titration.

The curve (**fig C**) is the same as that for a strong acid–strong base up to the equivalence point, but there is a difference once the acid is present in excess. Past the equivalence point, the solution contains a mixture of ethanoic acid and sodium ethanoate. This mixture acts as a buffer solution and therefore resists any large change in pH upon addition of further acid. (See **Section 14B.2**).

Note that the pH at the equivalence point is between 8 and 9; it is *not* 7.

TITRATION OF A STRONG ACID WITH A WEAK BASE

For this example we will add 1.00 mol dm^{-3} hydrochloric acid to 25 cm^3 of 1.00 mol dm^{-3} aqueous ammonia.



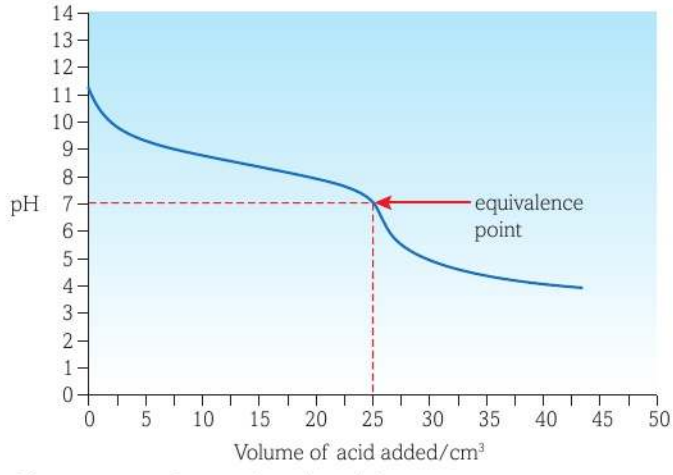
▲ **fig D** pH curve for a strong acid–weak base titration.

When the acid is first added the pH starts to fall quite sharply, but the curve quickly levels out (**fig D**). This is because a buffer solution has been formed, containing ammonia and ammonium chloride (again, see **Section 14B.2**).

Notice that the pH at the equivalence point is less than 7 because the salt formed, ammonium chloride, is composed of a strong acid and a weak base.

TITRATION OF A WEAK ACID WITH A WEAK BASE

For this example we will add 1.00 mol dm^{-3} ethanoic acid to 25 cm^3 of 1.00 mol dm^{-3} aqueous ammonia.



▲ **fig E** pH curve for a weak acid–weak base titration.

Notice that there is not any steep section to this graph (**fig E**). Instead, there is what is known as a 'point of inflection'. The lack of a steep section means that it is difficult to do a titration of a weak acid against a weak base using an indicator. The reason for this will be explained in **Section 14B.3**.

ACID–BASE INDICATORS

Earlier in this section we looked at the pH titration curves obtained when an aqueous solution of an acid is added to an aqueous solution of a base. The four cases we considered were:

- strong acid–strong base
- weak acid–strong base
- strong acid–weak base
- weak acid–weak base.

We are now going to use these four curves to help us understand why different indicators are required for different types of titrations.

An acid–base indicator is either a weak acid or a weak base. Most indicators are weak acids (HIn). For an indicator that is a weak acid, its dissociation in aqueous solution can be represented as:

$$HIn(aq) \rightleftharpoons H^+(aq) + In^-(aq)$$

The molecule, HIn, and its conjugate base, In^-, have different colours in aqueous solution. For methyl orange, these are red and yellow, respectively:

$$\underset{\text{red}}{HIn(aq)} \rightleftharpoons H^+(aq) + \underset{\text{yellow}}{In^-(aq)}$$

When $[H^+(aq)]$ is sufficiently large the equilibrium will shift far enough to the left for the red colour to predominate. If $[H^+(aq)]$ is very low then the equilibrium will lie far over to the right and the yellow colour will predominate. Therefore, the indicator changes colour according to the pH of the solution.

There will be a stage at which [HIn(aq)] = [In^-(aq)] and the indicator will appear orange. The exact pH at which this stage is reached can be determined using the equilibrium constant, K_{In}, for methyl orange.

$$K_{In} = \frac{[H^+(aq)][In^-(aq)]}{[HIn(aq)]} = 2.00 \times 10^{-4}\,mol\,dm^{-3}$$

When [HIn(aq)] = [In^-(aq)], the expression becomes:

$$[H^+(aq)] = 2.00 \times 10^{-4}\,mol\,dm^{-3}$$

This gives a pH of 3.70 for the 'half-way' stage. So, methyl orange will change colour at a pH of 3.70.

Note that this pH value is also the same as the value of pK_{In} for methyl orange. Therefore, the pH at which different indicators will change colour can be determined from their pK_{In} values.

pH RANGE OF INDICATORS

As a 'rule of thumb', the red colour of methyl orange will first predominate when [HIn(aq)] is ten times [In^-(aq)], and the yellow colour will predominate when [In^-(aq)] is ten times [HIn(aq)].

The approximate pH at which each colour predominates can be calculated as follows.

When [HIn(aq)] = 10[In^-(aq)]:

$$\frac{[H^+(aq)][In^-(aq)]}{10[In^-(aq)]} = 2.00 \times 10^{-4}\,mol\,dm^{-3}$$

$$[H^+(aq)] = 2.00 \times 10^{-3}\,mol\,dm^{-3}$$

So, the pH at which the red colour first predominates is 2.70.

A similar calculation will show that the pH at which the yellow colour first predominates is 4.70.

The 'pH range' of methyl orange is therefore approximately 2.70 to 4.70.

The *exact* pH range of methyl orange is 3.10 to 4.40. This means that at pH values below 3.10, methyl orange will appear red. At pH values above 4.40, methyl orange will appear yellow. Between 3.10 and 4.40, methyl orange will be a shade of orange.

Table A shows the pK_{In} values, pH ranges and colours of several common indicators.

INDICATOR	pK_{In}	pH RANGE	COLOUR	
			HIn(aq)	In^-(aq)
methyl orange	3.70	3.10–4.40	red	yellow
bromophenol blue	4.00	2.80–4.60	yellow	blue
bromothymol blue	7.00	6.00–7.60	yellow	blue
phenol red	7.90	6.80–8.40	yellow	red
phenolphthalein	9.30	8.20–10.00	colourless	red

table A

CHOICE OF INDICATOR

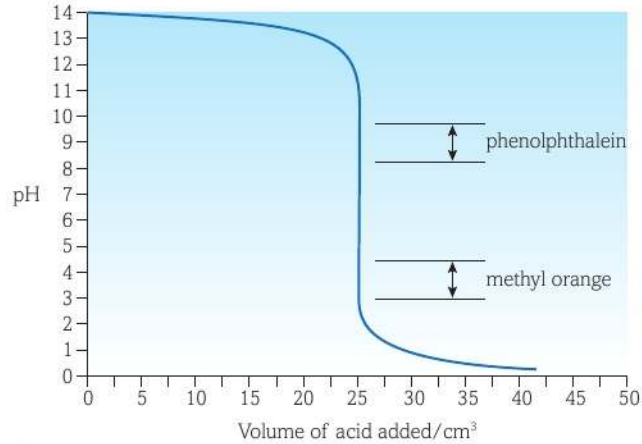
A good indicator shows a complete colour change upon the addition of one drop of acid from the burette. This is necessary in order to accurately determine the end point of the titration.

Therefore, there has to be a minimum pH change equivalent to the pH range of the indicator in order for the indicator to successfully determine the end point.

STRONG ACID–STRONG BASE TITRATION

Let us first of all consider the use of methyl orange and phenolphthalein as indicators for a strong acid–strong base titration.

Fig F shows the pH titration curve for 25 cm^3 of 1.00 mol dm^{-3} NaOH(aq) titrated with 1.00 mol dm^{-3} HCl(aq). The pH ranges of methyl orange and phenolphthalein have also been included.



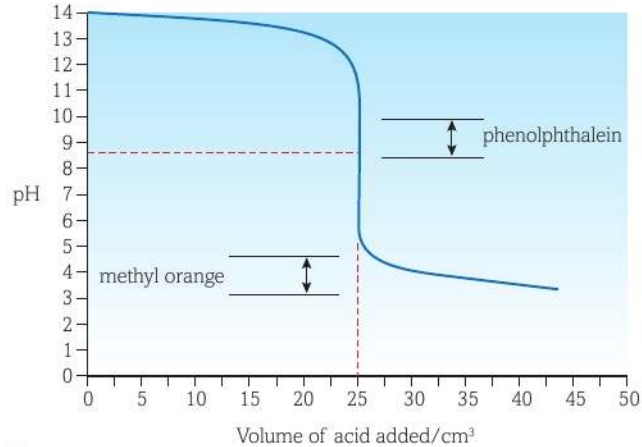
▲ **fig F** Strong acid–strong base pH curve.

You will notice that the pH range of each indicator falls within the steep section of the curve, where a large pH change is occurring upon the addition of just one drop of acid. This means that both indicators will change colour at the end point and therefore both are suitable indicators to use in this titration.

WEAK ACID–STRONG BASE TITRATION

We will now consider the suitability of each indicator for a weak acid–strong base titration.

Fig G shows the pH titration curve for 25 cm^3 of 1.00 mol dm^{-3} NaOH(aq) titrated with 1.00 mol dm^{-3} CH_3COOH(aq). Once again, the pH ranges of methyl orange and phenolphthalein have also been included.

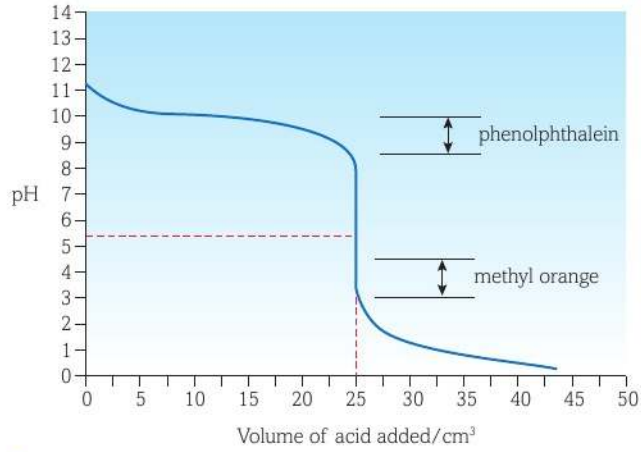


▲ **fig G** Weak acid–strong base pH curve.

This time, only the pH range of phenolphthalein falls within the steep section of the curve. So, phenolphthalein is suitable, but methyl orange is not.

STRONG ACID–WEAK BASE TITRATION

Fig H shows the pH titration curve for 25 cm^3 of 1.00 mol dm^{-3} $NH_3(aq)$ titrated with 1.00 mol dm^{-3} HCl(aq). Again, the pH ranges of methyl orange and phenolphthalein have also been included.

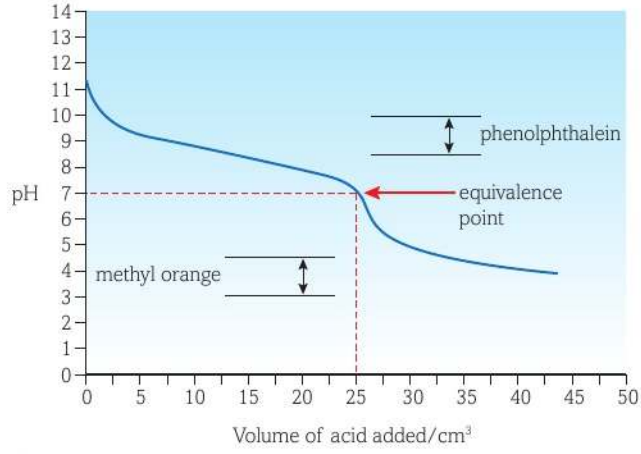


▲ **fig H** Strong acid–weak base pH curve.

This time, only the pH range of methyl orange falls within the steep section of the curve. So, methyl orange is suitable, but phenolphthalein is not.

WEAK ACID–WEAK BASE TITRATION

Fig I shows the pH titration curve for 25 cm^3 of 1.00 mol dm^{-3} $NH_3(aq)$ titrated with 1.00 mol dm^{-3} $CH_3COOH(aq)$. Again, the pH ranges of methyl orange and phenolphthalein have also been included.



▲ **fig I** Weak acid–weak base pH curve.

As there is no prominent steep section to the curve, neither indicator is suitable.

In fact, the end point of a titration of a weak acid and a weak base cannot be determined using an acid–base indicator. The end point of such a titration is best determined by measuring the temperature changes (thermometric titration) or electrical conductivity changes (conductometric titration).

TITRATION CURVES WITH DIPROTIC ACIDS

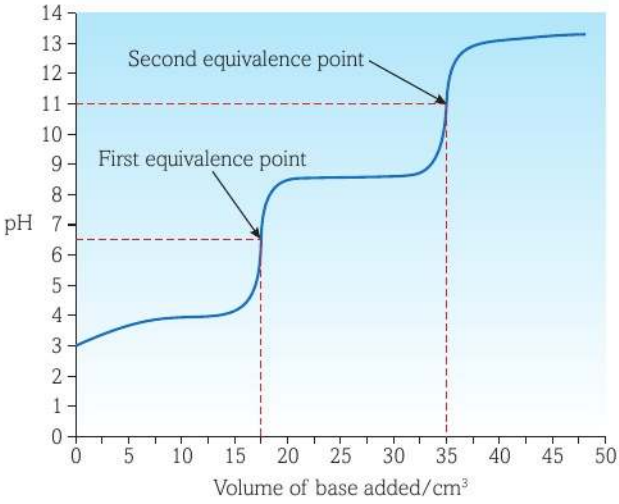
A diprotic acid is an acid that produces two H^+ ions per acid molecule. Examples of diprotic acids are sulfuric acid, H_2SO_4, and carbonic acid, H_2CO_3.

A diprotic acid dissociates in water in two stages:

Stage 1: $H_2X(aq) \rightarrow H^+(aq) + HX^-(aq)$

Stage 2: $HX^-(aq) \rightarrow H^+(aq) + X^{2-}(aq)$

Because of the successive dissociations (one following immediately after the other), the titration curves of diprotic acids have two equivalence points, as shown in **fig J**.



▲ **fig J** Diprotic acid–strong base pH curve.

The equations for the acid–base reactions occurring between a diprotic acid, H_2X, and the base sodium hydroxide, NaOH, are as follows.

From the beginning of the reaction to the first equivalence point:

$$H_2X + NaOH \rightarrow NaHX + H_2O$$

From the first equivalence point to the second equivalence point:

$$NaHX + NaOH \rightarrow Na_2X + H_2O$$

From the beginning of the reaction through to the second equivalence point (overall reaction):

$$H_2X + 2NaOH \rightarrow Na_2X + 2H_2O$$

At the first equivalence point, all H^+ ions from the first dissociation have reacted with NaOH.

At the second equivalence point, all H^+ ions from *both* reactions have reacted (twice as many as at the first equivalence point).

Therefore, the volume of NaOH added at the second equivalence point is exactly twice that at the first equivalence point.

CHOOSING THE BEST INDICATOR

The best indicator to choose for a particular titration is the one whose pK_{In} value is as close as possible to the pH at the equivalence point.

Bromothymol blue (pK_{In} = 7.00) is a particularly good indicator for a strong acid–strong base titration.

EXAM HINT

In an exam, you will be given a data book that contains the pK_{In} values, pH ranges and colours of 10 common indicators.

CHECKPOINT

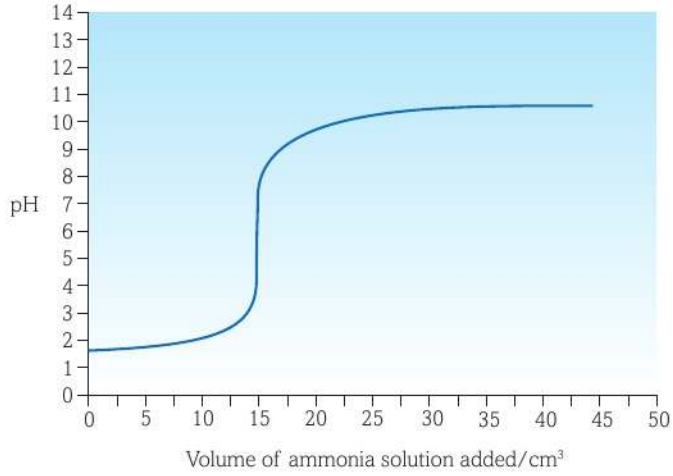
SKILLS INTERPRETATION

1. The equation for the reaction between hydrochloric acid and ammonia is:

$$HCl(aq) + NH_3(aq) \rightarrow NH_4Cl(aq)$$

A 25.0 cm^3 sample of 0.0200 mol dm^{-3} HCl(aq) was placed in a conical flask. Aqueous ammonia was added gradually from a burette and the pH was measured after each addition, until the pH no longer changed.

The pH curve for this titration is shown below.



(a) State how the curve suggests that ammonia is a weak base.

(b) Use the information given to calculate the concentration of the ammonia solution.

(c) The pH ranges for three indicators are:

Thymol blue: 1.2 to 2.8

Methyl red: 4.2 to 6.3

Thymolphthalein: 9.3 to 10.5

Explain which of these three indicators is the most suitable for this titration.

SKILLS INTERPRETATION

2. (a) Calculate the pH at 298 K of an aqueous solution of CH_3COOH of concentration 0.100 mol dm^{-3}. [$K_a = 1.74 \times 10^{-5}$ mol dm^{-3} at 298 K]

(b) Sketch the pH titration curve for the addition of 50.0 cm^3 of 0.100 mol dm^{-3} NaOH(aq) to 25.0 cm^3 of 0.100 mol dm^{-3} CH_3COOH(aq).

(c) State two differences in the pH curve that would be obtained if the titration were repeated using 25.0 cm^3 of 0.0500 mol dm^{-3} CH_3COOH(aq) instead of 25.0 cm^3 of 0.100 mol dm^{-3} CH_3COOH(aq).

SUBJECT VOCABULARY

equivalence point the point in a titration when the acid and base have reacted together in the exact proportions as dictated by the stoichiometric equation

14B 2 BUFFER SOLUTIONS

SPECIFICATION REFERENCE
14.17 14.18 14.19 14.20 14.22

LEARNING OBJECTIVES

- Know what is meant by the term 'buffer solution'.
- Understand the action of a buffer solution.
- Be able to calculate the pH of a buffer solution given appropriate data.
- Be able to calculate the concentrations of solutions required to prepare a buffer solution of a given pH.
- Understand the importance of buffer solutions in biological environments:
  (i) buffers in cells and in blood (H_2CO_3/HCO_3^-)
  (ii) in foods to prevent deterioration due to pH change (caused by bacterial or fungal activity).

WHAT IS A BUFFER SOLUTION?

Many experiments, particularly in biochemistry, have to be performed in aqueous solutions of fairly constant pH. Unfortunately, it is impossible to make a solution whose pH is totally unaffected by the addition of even small amounts of acid or base. However, it is possible to make a solution whose pH remains *almost* unchanged when small amounts of acid or base are added. Such a solution is called a **buffer solution**.

There are many ways of making a buffer solution, but two of the most common are:

- to mix a weak acid with its conjugate base
- to mix a weak base with its conjugate acid.

BUFFER MADE FROM A WEAK ACID AND ITS CONJUGATE BASE

The simplest example of this is ethanoic acid and sodium ethanoate. The salt of the weak acid has to be soluble in water, which is why sodium and potassium salts are commonly used to make buffer solutions.

In this mixture the acid is partially dissociated, whereas the salt is fully dissociated. The relevant equations are:

$$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$$

$$CH_3COONa(aq) \rightarrow CH_3COO^-(aq) + Na^+(aq)$$

This mixture will produce a buffer solution with a pH less than 7. The exact pH depends on the concentration of both acid and its conjugate base, and can be calculated as follows.

CALCULATING THE pH OF A BUFFER SOLUTION

For our example we will use a solution that has been made by mixing equal volumes of 1.00 mol dm^{-3} ethanoic acid and 1.00 mol dm^{-3} sodium ethanoate at 298 K.

If we assume that the extent of dissociation of the acid is negligible, then the concentration of CH_3COOH at equilibrium will be 0.500 mol dm^{-3}.

Also, again because the dissociation of the acid is negligible, the concentration of the ethanoate ions at equilibrium can be considered to be entirely made up from the sodium salt. So, the concentration of ethanoate ions at equilibrium is also 0.500 mol dm^{-3}.

Note that the concentration of both acid and salt in the mixture is half of the concentrations used because equal volumes of each solution were mixed. The total volume of the mixture is twice that of the original volume of each solution used.

At 298 K,

$$K_a = \frac{[CH_3COO^-(aq)][H^+(aq)]}{[CH_3COOH(aq)]} = 1.74 \times 10^{-5}\ \text{mol dm}^{-3}$$

Rearranging this equation and substituting values for $[CH_3COOH(aq)]$ and $[CH_3COO^-(aq)]$ gives:

$$[H^+(aq)] = (1.74 \times 10^{-5} \times 0.500)/0.500\ \text{mol dm}^{-3}$$
$$= 1.74 \times 10^{-5}\ \text{mol dm}^{-3}$$

This gives a pH for the buffer solution of 4.76.

LEARNING TIP

A potential trap when calculating the pH of a buffer solution is not understanding that when two solutions are mixed there is a dilution. If 50 cm^3 of one solution is mixed with 50 cm^3 of another solution, then the total volume of the mixture will be 100 cm^3. So, the concentration of each solution will be halved on mixing.

HOW DOES THE BUFFER ACTION WORK?

When a small amount of acid is added to the buffer solution, the majority of the H^+ ions added react with the CH_3COO^- ions to form CH_3COOH molecules:

$$CH_3COO^-(aq) + H^+(aq) \rightarrow CH_3COOH(aq)$$

When a little base is added, the majority of the OH^- ions added react with the CH_3COOH molecules:

$$CH_3COOH(aq) + OH^-(aq) \rightarrow CH_3COO^-(aq) + H_2O(l)$$

A new equilibrium mixture will be established in which the concentrations of both CH_3COOH and CH_3COO^- will have changed slightly from their original values. So, there will be a change in pH, but this will be minimal. This is shown by the following argument.

To show that the pH has changed very little, we once again need to make use of the expression for K_a of the acid:

$$K_a = \frac{[CH_3COO^-(aq)][H^+(aq)]}{[CH_3COOH(aq)]}$$

Rearranging this equation gives:

$$[H^+(aq)] = K_a \times \frac{[CH_3COOH(aq)]}{[CH_3COO^-(aq)]}$$

As there are a relatively large number of CH_3COOH molecules in the solution, as the extent of dissociation of the acid is very small, the change in $[CH_3COOH(aq)]$ will be negligible.

Also, the change in $[CH_3COO^-(aq)]$ will be negligible because there are a relatively large number of CH_3COO^- ions present, resulting from the total dissociation of the CH_3COONa.

This means that the ratio $\frac{[CH_3COOH(aq)]}{[CH_3COO^-(aq)]}$ will remain fairly constant.

If this ratio remains fairly constant, then $[H^+(aq)]$ remains fairly constant because, at a given temperature, K_a is also constant.

If $[H^+(aq)]$ remains fairly constant, then the pH remains fairly constant.

To summarise, a buffer solution of a weak acid and its conjugate base maintains a fairly constant pH because the ratio of $[CH_3COOH(aq)]$ to $[CH_3COO^-(aq)]$ remains fairly constant when *small* amounts of either acid or base are added.

The best way to demonstrate the effect of adding a small amount of acid on the pH of a buffer solution is to calculate the pH of the buffer both before and after adding the acid.

We have already calculated the pH of a buffer solution made by mixing equal volumes of 1.00 mol dm^{-3} ethanoic acid and 1.00 mol dm^{-3} sodium ethanoate at 298 K. It is 4.76.

Let us imagine that we have mixed 500 cm^3 of each solution to make 1 dm^3 of buffer solution. The amounts of CH_3COOH and CH_3COO^- in this solution will both be equal to 0.500 mol.

Let us imagine we have 1 dm^3 of this solution and we add 1.00×10^{-2} (0.0100) mol of HCl to it. To make the mathematics easier we will assume that the volume of the solution does not change.

0.0100 mol of HCl will provide 0.0100 mol of H^+ ions. These will react with the CH_3COO^- ions in the buffer in a 1 : 1 molar ratio:

$$CH_3COO^-(aq) + H^+(aq) \rightarrow CH_3COOH(aq)$$

The amount of CH_3COOH present will now have increased from 0.500 to 0.510 mol.

The amount of CH_3COO^- present will have decreased from 0.500 to 0.490 mol.

So, the new concentrations of acid and base present are:

$[CH_3COOH(aq)] = 0.510 \text{ mol dm}^{-3}$

and $[CH_3COO^-(aq)] = 0.490 \text{ mol dm}^{-3}$

The new hydrogen ion concentration is given by:

$$[H^+(aq)] = 1.74 \times 10^{-5} \times \frac{0.510}{0.490} = 1.81 \times 10^{-5} \text{ mol dm}^{-3}$$

The new pH = $-\lg (1.81 \times 10^{-5}) = 4.74$ (to 3 significant figures).

The pH has changed by 0.02 units from 4.76 to 4.74.

To understand how effective the buffer solution is in controlling the pH, let us consider adding 0.0100 mol of H^+ ions to 1 dm^3 of deionised water.

The pH, at 298 K, of deionised water is 7.00 so:

$$[H^+(aq)] = 1.00 \times 10^{7} \text{ mol dm}^{-3}$$

Adding 0.0100 mol of H^+ ions gives:

$$[H^+(aq)] = (1.00 \times 10^{-7} + 0.0100) = 1.00001 \times 10^{-2} \text{ mol dm}^{-3}$$

The new pH is given by:

$$\text{pH} = -\lg (1.00001 \times 10^{-2}) = 2.00$$

The pH has dropped by 5 units as opposed to 0.02 units with the buffer solution. A considerable difference!

HENDERSON–HASSELBALCH EQUATION

The Henderson–Hasselbalch equation can also be used to calculate the pH of a buffer solution.

For a weak acid, HA, and its conjugate base, A^-, the following equation applies:

$$[H^+(aq)] = K_a \times \frac{[HA(aq)]}{[A^-(aq)]}$$

Strictly speaking, the concentration terms in the equation are the concentrations at equilibrium. However, it is reasonable to take the original concentrations, for reasons we have already discussed.

The original concentration of A^- will be the same as that of the salt, provided that a sodium or potassium salt has been used. The equation can now be rewritten as:

$$[H^+(aq)] = K_a \times \frac{[\text{acid}]}{[\text{salt}]}$$

If we take the logarithm to base 10 of both sides of this equation we get:

$$\lg [H^+(aq)] = \lg K_a + \lg \frac{[\text{acid}]}{[\text{salt}]}$$

Or:

$$-\lg [H^+(aq)] = -\lg K_a - \lg \frac{[\text{acid}]}{[\text{salt}]}$$

Or:

$$\text{pH} = \text{p}K_a - \lg \frac{[\text{acid}]}{[\text{salt}]}$$

Or:

$$\text{pH} = \text{p}K_a + \lg \frac{[\text{salt}]}{[\text{acid}]}$$

This last equation is the most common form of the Henderson–Hasselbalch equation.

LEARNING TIP

Do not worry if you did not fully understand the derivation of the Henderson–Hasselbalch equation. You do not need to learn this equation for the exam, although you may use it if you wish. The equation is also given in another form in which [base] replaces [salt], because A^- is the base in the buffer.

If we use the Henderson–Hasselbalch equation to calculate the pH of our solution containing ethanoic and ethanoate ions, both of concentration 0.500 mol dm^{-3}, we obtain:

$$\text{pH} = 4.76 + \lg (0.500/0.500) = 4.76$$

$[\text{p}K_a = -\lg K_a = -\lg (1.74 \times 10^{-5})$; $\lg 1 = 0]$

This is the same answer as we obtained earlier.

BUFFER MADE FROM A WEAK BASE AND ITS CONJUGATE ACID

The most common example of this type of buffer is ammonia and the ammonium ion. The ammonium ion is usually supplied in the form of ammonium chloride. This mixture will provide a buffer solution with a pH greater than 7.

The most convenient equilibrium to consider is:

$$NH_4^+(aq) \rightleftharpoons NH_3(aq) + H^+(aq)$$

This mixture provides a relatively high concentration of both NH_3 molecules and NH_4^+ ions.

The buffer works in a similar manner to the weak acid–conjugate base system. The addition of acid results in the added H^+ ions reacting with NH_3 molecules:

$$NH_3(aq) + H^+(aq) \rightarrow NH_4^+(aq)$$

whereas the addition of base results in the added OH^- ions reacting with NH_4^+ ions:

$$NH_4^+(aq) + OH^-(aq) \rightarrow NH_3(aq) + H_2O(l)$$

As there is a relatively high concentration of both NH_3 molecules and NH_4^+ ions, the ratio of $[NH_3(aq)]$ to $[NH_4^+(aq)]$ remains relatively constant when we add small amounts of either acid or base. This results in the pH remaining fairly constant because it is given by the equation:

$$pH = pK_a + \lg \frac{[NH_3(aq)]}{[NH_4^+(aq)]}$$

As pK_a is constant at a given temperature, the pH of the solution depends on the ratio of $[NH_3(aq)]$ to $[NH_4^+(aq)]$.

HOW TO MAKE A BUFFER SOLUTION WITH A REQUIRED pH

To make a buffer solution with a pH less than 7, you need to use a mixture of a weak acid and its conjugate base.

In the opposite way, you need to use a mixture of a weak base and its conjugate acid to make a buffer solution with a pH greater than 7.

WORKED EXAMPLE 1

Imagine we have to make a buffer solution of pH 5.00 at a temperature of 298 K.

To make this solution, we need a hydrogen ion concentration, $[H^+(aq)]$, of 1.00×10^{-5} mol dm^{-3}.

The hydrogen ion concentration of a buffer solution of a weak acid and its conjugate base is calculated using the formula:

$$[H^+(aq)] = K_a \times \frac{[acid]}{[salt]}$$

If we use ethanoic acid as the weak acid, then:

$$K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$$

If we now substitute our known values into the equation we obtain:

$$1.00 \times 10^{-5} = 1.74 \times 10^{-5} \times \frac{[acid]}{[salt]}$$

This gives a value for:

$$\frac{[acid]}{[salt]} \text{ of } (1.00 \times 10^{-5} \div 1.74 \times 10^{-5}) = 0.575$$

(to 3 significant figures)

So, if we were supplied with an ethanoic acid solution of concentration 0.575 mol dm^{-3} and a sodium ethanoate solution of 1.00 mol dm^{-3}, we could make a buffer solution of pH 5.00 by mixing equal volumes of the two solutions. This would give us a solution in which the acid concentration was 0.2875 mol dm^{-3} and the salt concentration was 0.500 mol dm^{-3}.

[(0.2875 ÷ 0.500) = 0.575]

WORKED EXAMPLE 2

In what proportions should we mix 0.100 mol dm^{-3} solutions of ammonia and ammonium chloride to obtain a buffer of pH 9.80?

The equation for the dissociation of NH_4^+ in aqueous solution is:

$$NH_4^+(aq) \rightleftharpoons NH_3(aq) + H^+(aq)$$

$$K_a = \frac{[NH_3(aq)][H^+(aq)]}{[NH_4^+(aq)]} = 5.62 \times 10^{-10} \text{ mol dm}^{-3}$$

$$\frac{[NH_3(aq)]}{[NH_4^+(aq)]} = \frac{5.62 \times 10^{-10}}{[H^+(aq)]}$$

$$[H^+(aq)] = 10^{-pH} = 10^{-9.80} = 1.58 \times 10^{-10} \text{ mol dm}^{-3}$$

$$\frac{[NH_3(aq)]}{[NH_4^+(aq)]} = \frac{5.62 \times 10^{-10}}{1.58 \times 10^{-10}} = 3.56$$

Therefore, we have to mix the solutions in a ratio by volume of 3.56 $NH_3(aq)$ to 1 $NH_4Cl(aq)$.

CONTROLLING THE pH OF BLOOD

The human body works within a narrow range of pH values. For example, the pH of arterial blood plasma needs to be in the range of 7.35 to 7.45. The way that the whole body functions would be affected if the pH of this blood plasma were to change significantly, particularly if it were to fall.

The pH of blood in cells is controlled by a mixture of buffers. The most important one is the carbonic acid–hydrogen carbonate buffer mixture.

In this mixture the carbonic acid molecule, H_2CO_3, acts as the weak acid. The hydrogen carbonate ion, HCO_3^-, is the conjugate base of H_2CO_3.

The equilibrium that exists is represented by the equation:

$$H_2CO_3(aq) \rightleftharpoons HCO_3^-(aq) + H^+(aq)$$

Under normal circumstances, the amount of HCO_3^- ion present is approximately 20 times that of H_2CO_3. As normal metabolism produces more acids than bases, this is consistent with the needs of the body.

Any increase in the concentration of hydrogen ions in the blood (for example, by the production of lactic acid in the muscles), results in the equilibrium shown above moving to the *left* as the added H^+ ions react with the HCO_3^- ions.

A variety of respiratory and metabolic factors can also cause the pH of the blood to rise. For example, the overuse of diuretics increases the amount of urine excreted from the body. If the urine contains large amounts of acids, then the pH of the blood will increase. If this happens, the equilibrium shown above will move to the *right* as the H_2CO_3 molecules ionise to increase the H^+ concentration and restore the pH to its normal level.

This is considered to be the most important buffer because it is coupled with the respiratory system of the body. Carbonic acid is not particularly stable, and in aqueous solution it decomposes to form carbon dioxide and water:

$$H_2CO_3(aq) \rightleftharpoons CO_2(aq) + H_2O(aq)$$

It is the respiratory system that is responsible for removing carbon dioxide from the body. Aqueous carbon dioxide exists in equilibrium with gaseous carbon dioxide:

$$CO_2(g) \rightleftharpoons CO_2(aq)$$

Combining these three reactions gives us:

$$CO_2(g) + H_2O(aq) \rightleftharpoons CO_2(aq) + H_2O(aq) \rightleftharpoons H_2CO_3(aq) \rightleftharpoons HCO_3^-(aq) + H^+(aq)$$

When these equilibria shift to the left as a result of an increase in hydrogen ion concentration, the concentration of carbon dioxide in the blood increases. The carbon dioxide leaves the blood in the lungs and is then exhaled, thus maintaining the normal pH of the blood.

BUFFERS IN FOOD

A combination of various factors such as light, oxygen, heat, humidity and/or many kinds of microorganisms (bacteria and fungi) can spoil food. We try to reduce spoilage by keeping certain foods in the dark, in airtight containers and/or in refrigerators.

Spoilage of food by microorganisms depends greatly on the pH value of the food. Most microorganisms thrive when the pH of their surroundings is close to neutral (pH 6.6–7.5). The metabolism of these microorganisms is then greatest and they can multiply quickly. Most bacteria can survive at pH values as low as 4.4 and as high as 9.0. Only specialised bacteria can survive outside this range.

One important factor in the spoilage of food is its **buffer capacity**. Buffer capacity is a measure of the amount of acid or base required to change significantly the pH of the food. The more protein there is in the food, the higher is its buffer capacity. This is because the amino acids present have both acidic and basic properties. This means that it takes longer for the pH of the food to change enough for the bacteria to start multiplying. Bacteria and moulds can also produce waste products that act as poisons and toxins, causing ill-effects.

DID YOU KNOW?

Many processed foods, such as jams, contain buffer systems such as citric acid and sodium citrate. These help to maintain the pH within a range where the growth of microorganisms is very slow or non-existent.



▲ **fig A** Jam is an example of a processed food that can contain citric acid and sodium citrate.

CHECKPOINT

1. (a) Explain what is meant by the term 'buffer solution'.

(b) Explain how an aqueous solution containing a mixture of methanoic acid, HCOOH, and potassium methanoate, HCOOK, acts as a buffer.

(c) A buffer solution contains equal concentrations of methanoic acid and potassium methanoate.
Explain the effect on the pH of this solution of adding some solid potassium methanoate.

(d) Calculate the pH, at 298 K, of a buffer solution made by mixing equal volumes of 1.00 mol dm^{-3} methanoic acid and 0.500 mol dm^{-3} potassium methanoate.
[K_a(HCOOH) = 1.79×10^{-4} mol dm^{-3}]

SKILLS CREATIVITY

2. A student prepares two solutions.

Solution **A** is prepared by mixing 50 cm^3 of 0.100 mol dm^{-3} CH_3COOH(aq) with 25 cm^3 of 0.100 mol dm^{-3} NaOH(aq).

Solution **B** is prepared by mixing 25 cm^3 of 0.200 mol dm^{-3} CH_3COOH(aq) with 50 cm^3 of 0.100 mol dm^{-3} NaOH(aq).

Explain why solution **A** is a buffer solution but solution **B** is not.

3. A buffer solution was made by mixing 50 cm^3 of 0.200 mol dm^{-3} aqueous ammonia, NH_3(aq), with 50 cm^3 of aqueous ammonium chloride, NH_4Cl(aq). The pH of the resulting solution was 9.55. Calculate the concentration of the NH_4Cl(aq) used.
[$K_a(NH_4^+) = 5.62 \times 10^{-10}$ mol dm^{-3}]

4. Calculate the pH of a buffer solution containing 12.20 g of benzoic acid (C_6H_5COOH) and 7.20 g of sodium benzoate (C_6H_5COONa) in 1.00 dm^3 of solution.
[$pK_a(C_6H_5COOH) = 4.20$]

5. Like water, liquid ammonia undergoes self-dissociation:

$$2NH_3 \rightleftharpoons NH_4^+ + NH_2^-$$

(a) Explain why ammonia can be classified as an amphoteric substance.

(b) For each of the following substances, indicate whether a solution of it in liquid ammonia will be 'acidic', 'basic' or 'neutral'.
(i) Ammonium chloride, NH_4Cl
(ii) Sodium amide, $NaNH_2$
(iii) Potassium hydroxide, KOH

SUBJECT VOCABULARY

buffer solution a solution that *minimises* the change in pH when a *small* amount of either acid or base is added
buffer capacity a measure of the amount of acid or base required to change significantly the pH of food or of a solution of an acid and a base

14B 3 BUFFER SOLUTIONS AND pH CURVES

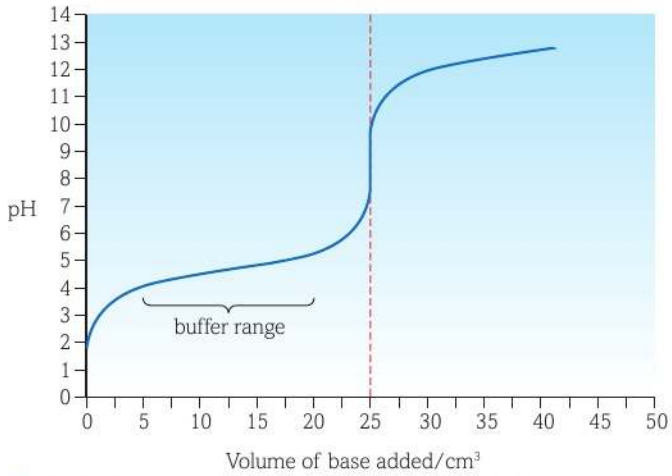
SPECIFICATION REFERENCE
14.21 CP11

LEARNING OBJECTIVES

- Understand how to use a weak acid–strong base or strong acid–weak base titration curve to:
  (i) demonstrate buffer action
  (ii) determine K_a from the pH at the point where half the acid is neutralised/equivalence point.

BUFFER ACTION DURING A TITRATION

Fig A shows a typical pH curve obtained when a weak acid is titrated against a strong base.



▲ **fig A** Strong base–weak acid pH curve showing buffer range.

EXAM HINT

When a weak acid is titrated against a strong base, there is a fairly rapid change of about 1.5 pH units when the strong base is first added. This is before the buffer range begins. Be sure to show this clearly if you are asked to **sketch** a curve for this titration in an exam question.

In the region marked 'buffer range', the change in pH as the base is added is gradual. Over this range there is a considerable concentration of both acid and conjugate base molecules. This mixture is displaying buffer action.

DETERMINING K_a FROM A pH TITRATION CURVE

This experiment involves performing a titration with an aqueous solution of a weak acid in a conical flask. From a burette, we add a standard solution of a strong base, such as sodium hydroxide. We measure the pH of the solution after each addition.

Then we are able to plot a graph of pH against volume of base added. From this graph we can determine the minimum volume of base required to completely react with all of the acid.

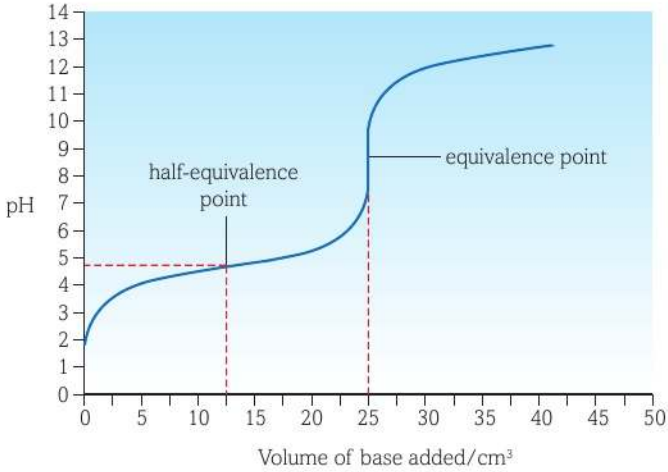
Then we can use the graph to determine the pH at the half-equivalence point. This pH value is equal to the pK_a value of the weak acid. From this it is a simple matter to calculate K_a for the acid.

A typical pH titration curve is shown in **fig B**.

The volume at the equivalence point is 25 cm^3, so the volume at the half-equivalence point is 12.5 cm^3. The pH when 12.5 cm^3 of base is added is 4.80.

The pK_a of the acid = 4.80

K_a for the acid = $10^{-4.80} = 1.58 \times 10^{-5}$ mol dm^{-3}



▲ **fig B** Strong base–weak acid pH curve showing equivalence and half-equivalence points.

EXAM HINT

Make sure you use the term half-equivalence point and *not* half-neutralisation point when determining the pK_a of a weak acid.

SUPPORTING THEORY

The theoretical justification for determining K_a for a weak acid by the method shown above is quite straightforward.

The mixture at the half-equivalence point is a buffer solution (as stated above). The pH of a buffer solution is calculated using the following equation:

$$pH = pK_a + \lg \frac{[salt]}{[acid]}$$

At the half-equivalence point:

[salt] = [acid], so [salt]/[acid] = 1

The logarithm to the base 10 of 1 (lg 1) = 0.

So, the equation becomes:

$$pH = pK_a$$

ALTERNATIVE METHOD

This method has been called the 'half-volume method'.

- Using a volumetric pipette, place 25.0 cm^3 of an aqueous solution of the weak acid into a conical flask.
- Add a few drops of phenolphthalein indicator.
- Titrate against a solution of aqueous sodium hydroxide until the end point colour is obtained.
- Note the volume of sodium hydroxide required. This is the minimum volume required to completely react with the acid.
- Use a fresh 25.0 cm^3 sample of the same aqueous solution of the weak acid and the same aqueous solution of sodium hydroxide, but this time do not add the phenolphthalein.
- Add only *half* the volume of sodium hydroxide required to react with the acid.
- Measure the pH of this solution. This pH value is equal to the pK_a value of the acid.

PRACTICAL SKILLS CP11

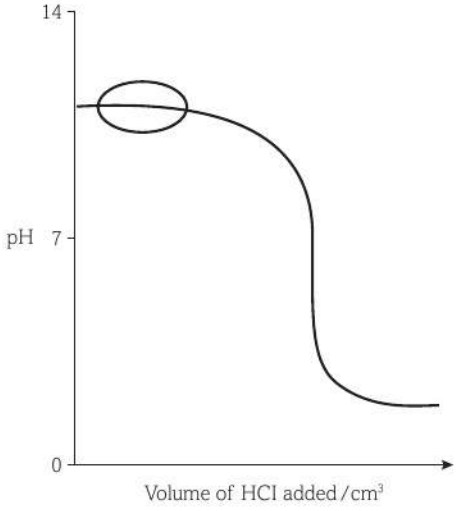
Note that you also considered this core practical in **Section 14A.4.**

Safety Note: Wear eye protection and avoid skin contact with the solutions. Make sure the top of the burette is below eye level when you fill it.

CHECKPOINT

1. A student carried out a titration by adding hydrochloric acid to ammonia solution. A sketch graph of pH against volume of hydrochloric acid added is shown below.

SKILLS INTERPRETATION



(a) Name the type of solution formed in the region ringed on the sketch graph.

(b) Explain how you deduced your answer to part (a).

2. The diagram shows the titration curve for a weak acid.

SKILLS INTERPRETATION

Use the curve to determine the pK_a value for the acid. Explain how you arrived at your answer.

